

Advanced Computer Networks Lab



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Lab Assignment-3

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**DSSRG: Decentralized
Smart Systems Research
Group**

<https://sites.google.com/iitbbs.ac.in/dssrg>

Or Google dssrg iitbbs



About

- Transport layer protocols: TCP vs UDP
- TCP is connection oriented, ensures data integrity through ordering, acknowledgement and retransmission.
- UDP is fast, connectionless, and unreliable protocol.
- TCP has everything, why UDP then?
- UDP gives blank slate. Implement additional functionality as required.
- This assignment focuses on such applications that can be created on top of UDP.



Learning outcome (A3.1 and 3.2)

- How to transfer binary data via sockets.
- Parsing binary from sockets to data structure.
- Video streaming application on top of UDP.
- How basic video streaming works.
- How to efficiently transfer data (video frames) over network.
- Make the connection reliable.



Assignment 3.1

Problem Statement:

Build a simple video streaming client for a video streaming server provided to you in executable form. The video streaming would be built on top of UDP. After running the server it will wait for client to get connected. The client when connected will first receive video file metadata, then it will receive all the frames. The frames would be sent one after another automatically, the client just have to read the frames and display them. You don't need to implement the video player GUI, you have to use the OpenCV library.

Download sample video from:

<https://www.peach.themazzone.com/durian/movies/sintel-1024-surround.mp4>



Assignment 3.1



The screenshot shows a terminal window with a dark background. The title bar at the top indicates the user is 'dell@DESKTOP-09BUBKJ' in the directory '/mr', with a 'Settings' button and window controls. The terminal content shows the user running 'build2/server' in the directory '/mnt/d/projects_wsl/acnlab-udp-video-streaming'. The output shows 'Video streaming server' and 'server: waiting to recvfrom...'. The terminal is split into two panes. The bottom status bar is green and shows '[0] 0: bash*' on the left and '"DESKTOP-09BUBKJ" 15:08 20-Mar-26' on the right.

```
dell@DESKTOP-09BUBKJ: /mnt/d/projects_wsl/acnlab-udp-video-streaming$ build2/server
Video streaming server
server: waiting to recvfrom...
```

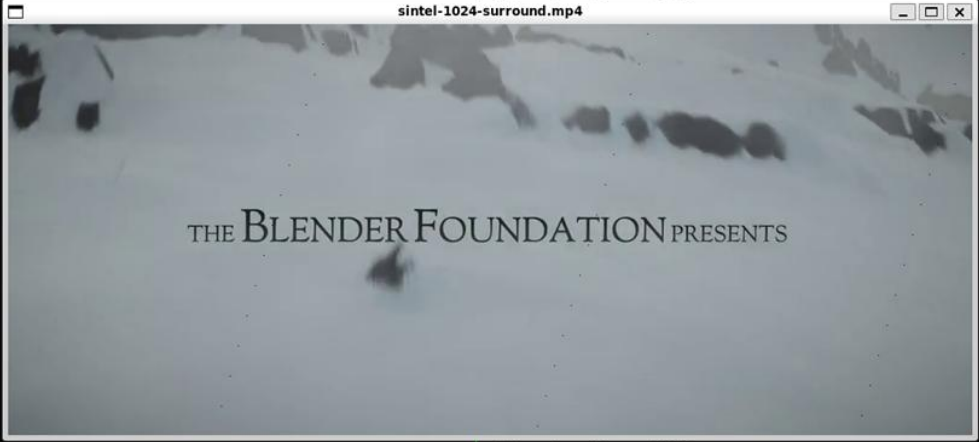


Assignment 3.1

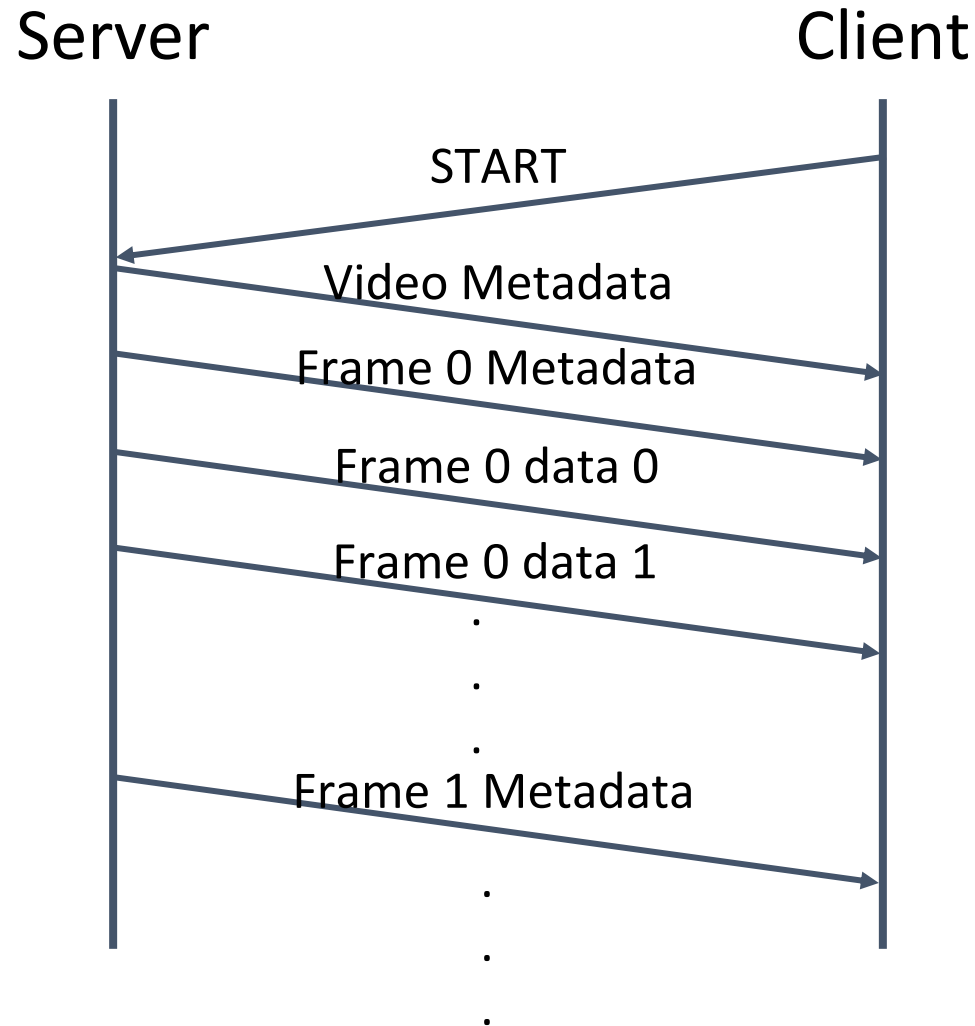
```
dell@DESKTOP-09BUBKJ: /mr  Settings  X + v

frame103 client: displaying image
frame104 client: frame135
frame105 client: frame135
frame106 client: frame135
frame107 client: frame135
frame108 client: frame135
frame109 client: frame135
frame110 client: frame135
frame111
frame112
frame113
frame114
frame115
frame116
frame117
frame118
frame119
frame120
frame121
frame122
frame123
frame124
frame125
frame126
frame127
frame128
frame129
frame130
frame131
frame132
frame133
frame134
frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135

[0] 0:build2/client* "DESKTOP-09BUBKJ" 15:13 20-Mar-26
```

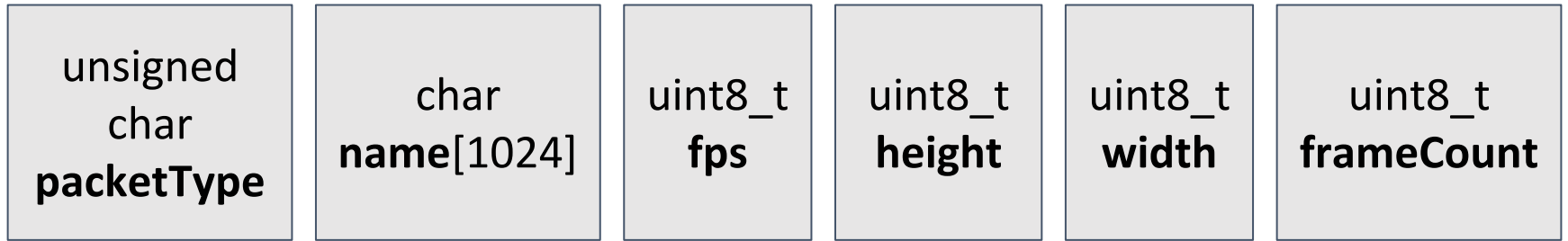


Assignment 3.1

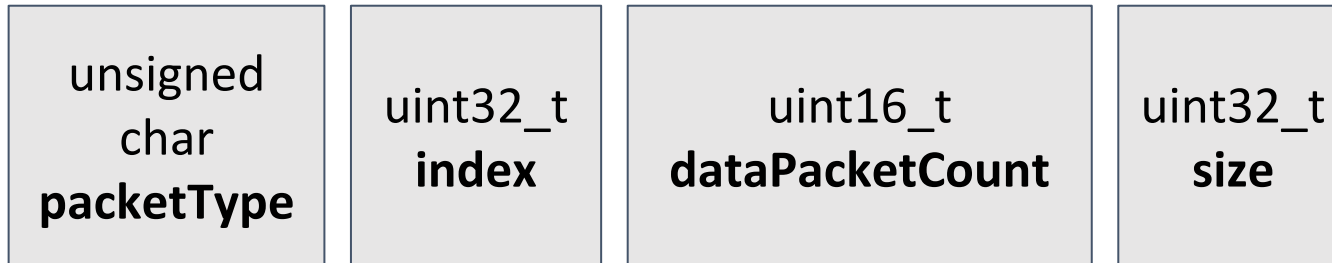


Assignment 3.1

Video metadata

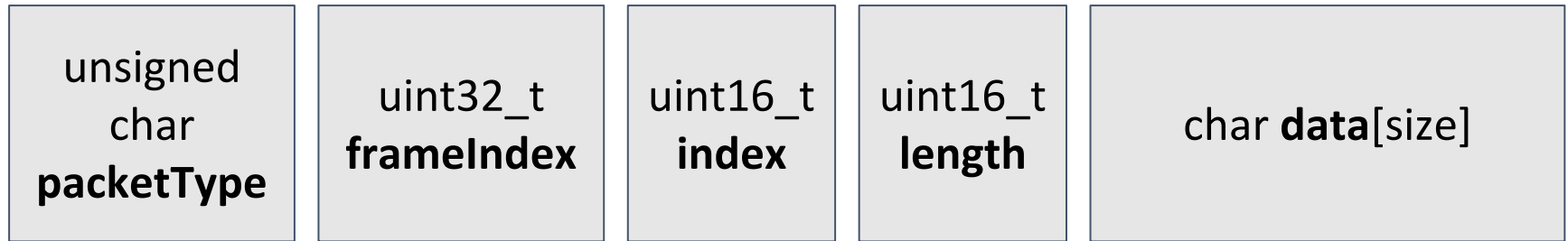


Frame metadata



Assignment 3.1

Frame data packet



*#define MAXBUFLLEN 32 * 1024*

size = MAXBUFLLEN - extra_size

*extra_size = sizeof(unsigned char) + sizeof(uint32_t)
+ 2 * sizeof(uint16_t)*

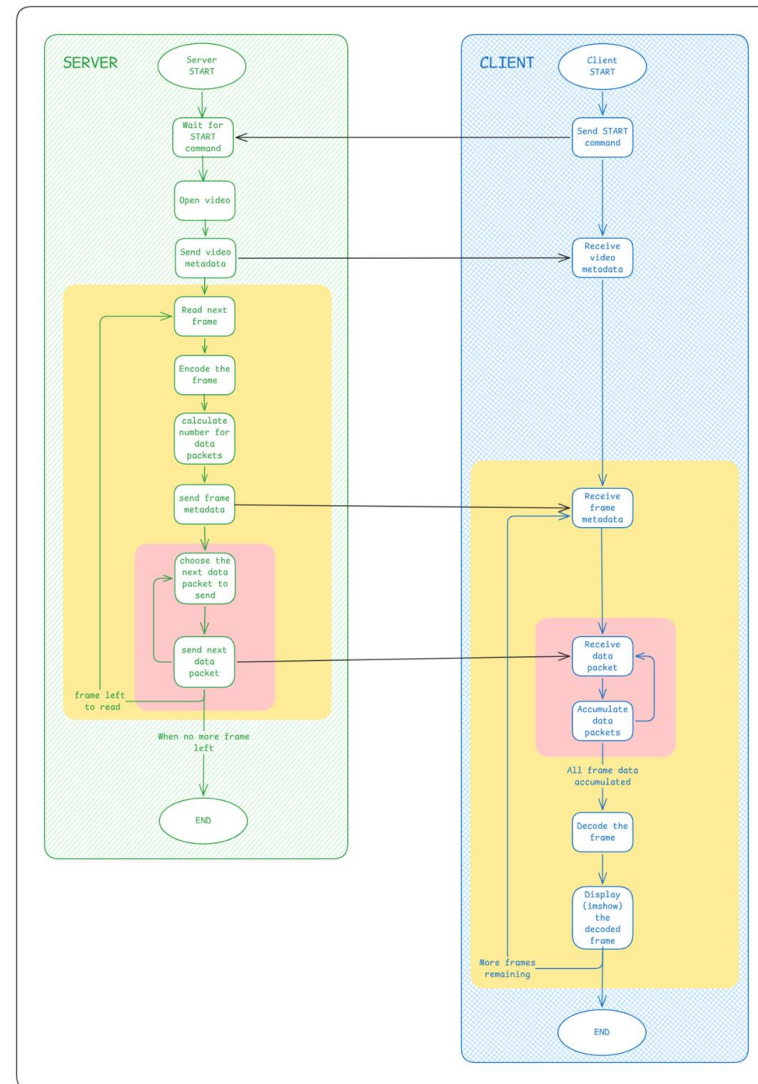
Assignment 3.1

packetType:

- 1 - Video metadata
- 2 - Frame metadata
- 3 - Frame data packet



Assignment 3.1



Assignment 3.1

```
dell@DESKTOP-09BUBKJ:/mnt/d/projects_wsl/acnlab-udp-reliable-video-streaming$ build/server -h
Video streaming server
udp video streaming server

server [OPTIONS]

OPTIONS:
-h,      --help                Print this help message and exit
-f,      --filename TEXT [sintel-1024-surround.mp4]
                               Video filename path
-p,      --port TEXT [8000]    Port of the server
-d,      --delay-between-frames INT [30]
                               Delay between sending frames (>0)
-q,      --encode-jpeg-quality INT [80]
                               JPEG quality while encoding each frames (0-100)
```



Assignment 3.1

Run the server:

- `build/server` will start the server with port 8000 and read and send the video sintel-1024-surround.mp4
- `build/server -p 8001` will start the server in different port (here 8001)
- `build/server -f [video_path]` will read a different video



Assignment 3.2

Problem Statement:

In previous assignment 2.1, you implemented a simple video streaming client. The channel was assumed to be noiseless, no packet would be dropped and they will be received in sequence. In assignment 2.2, you have to implement the video streaming client making sure that the video frames are received in order and dropped packets should be re-requested after a certain duration. Again the video streaming server would be provided in form of executable and this time you have to request for a frame rather than server sending them automatically.



Assignment 3.2

```
dell@DESKTOP-09BUBKJ: /mr  Settings  X + v

frame103
frame104
frame105
frame106
frame107
frame108
frame109
frame110
frame111
frame112
frame113
frame114
frame115
frame116
frame117
frame118
frame119
frame120
frame121
frame122
frame123
frame124
frame125
frame126
frame127
frame128
frame129
frame130
frame131
frame132
frame133
frame134
frame135

client: displaying image
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135

sintel-1024-surround.mp4

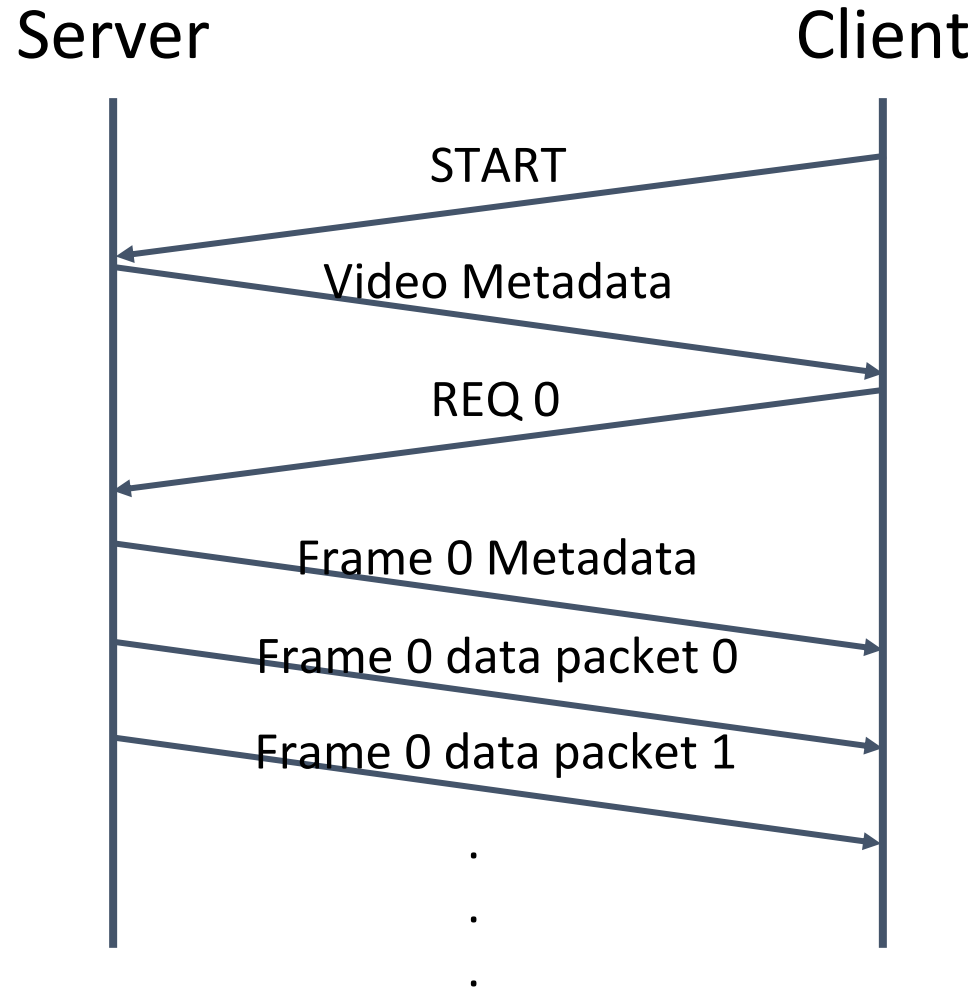
THE BLENDER FOUNDATION PRESENTS

client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135
client: frame135

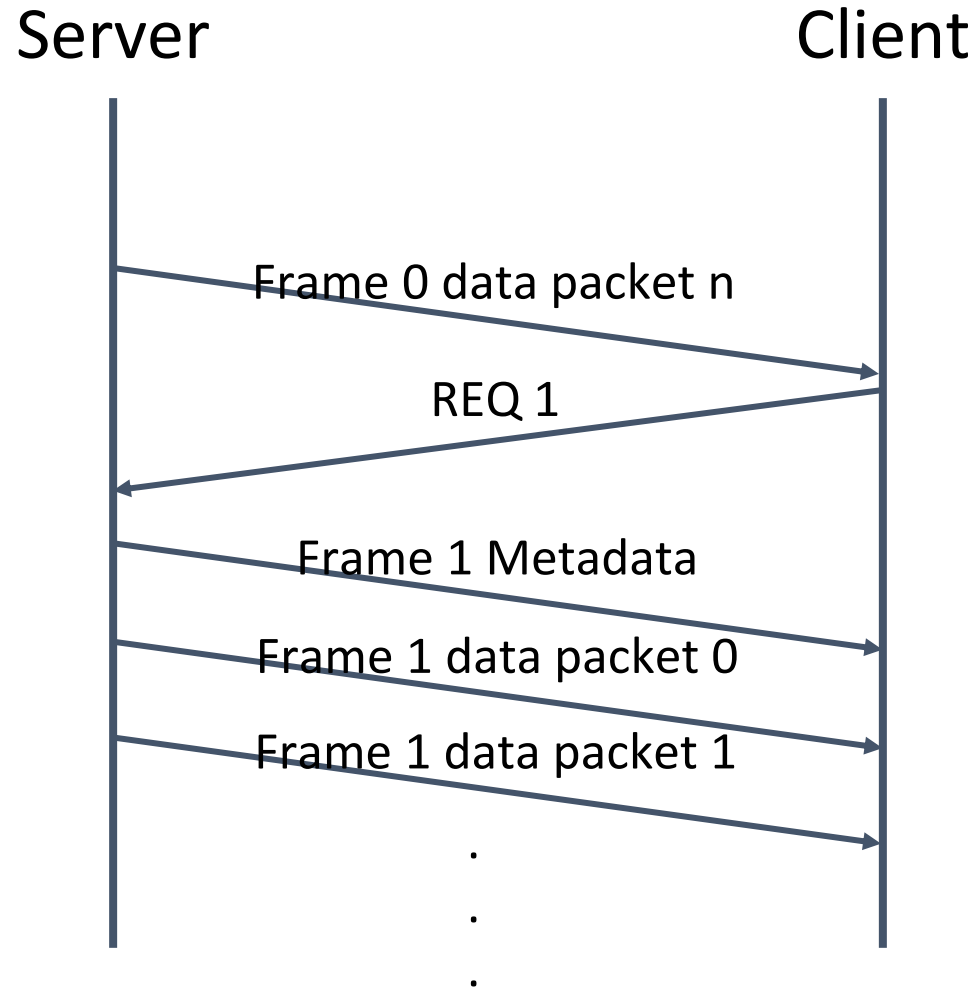
[0] 0:build2/client* "DESKTOP-09BUBKJ" 15:13 20-Mar-26
```

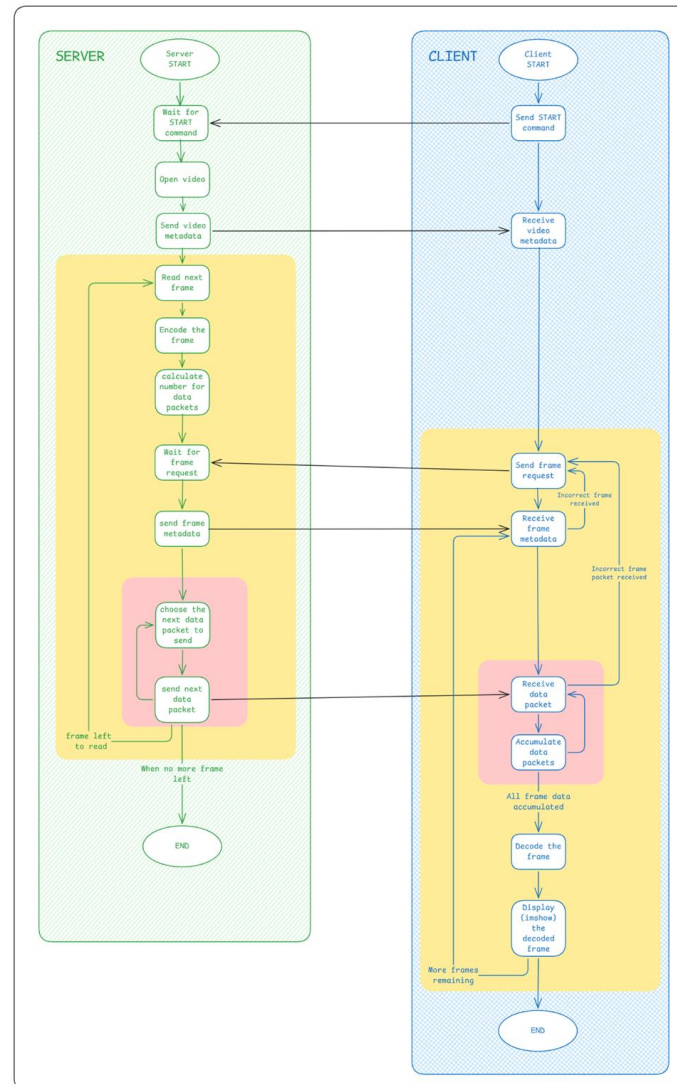


Assignment 3.2



Assignment 3.2





Assignment 3.2

```
dell@DESKTOP-09BUBKJ:/mnt/d/projects_wsl/acnlab-udp-reliable-video-streaming$ build/server2 -h
Video streaming server
udp video streaming server
```

```
server [OPTIONS]
```

OPTIONS:

```
-h,      --help                Print this help message and exit
-f,      --filename TEXT [sintel-1024-surround.mp4]
                               Video filename path
-p,      --port TEXT [8000]    Port of the server
-d,      --delay-between-frames INT [30]
                               Delay between sending frames (>0)
                               --packet-drop-prob FLOAT [0.1]
                               Packet drop probability (0-1)
-q,      --encode-jpeg-quality INT [80]
                               JPEG quality while encoding each frames (0-100)
-r,      --disable-random-packet-drop
                               Enable random packet drop
-s,      --fixed-seed          Fixed seed for random number generator
-c,      --continuous-packet-drop
                               Drop packet throughout the runtime.
```



Assignment 3.2

Run the server:

- `build/server2` will start the server with port 8000 and read and send the video sintel-1024-surround.mp4
- `build/server2 -p 8001` will start the server in different port (here 8001)
- `build/server2 -f [video_path]` will read a different video
- `build/server2 -r` to disable random packet drops
- `build/server2 -packet-drop-prob 0.3` to set the packet drop probability to 0.3



Thank you

